Problem Statement and Goals
Flexin

Team 3, Flexin
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Table 1: Revision History

Date	Developer(s)	Change
September 24th, 2026	Noah	Added goals
September 25th, 2026	Noah	Added stretch goals
September 25th, 2026	Peter	Added inputs, outputs and stakeholders; fixed hyphenation and clarified the fitness compounding habit goal
September 26th, 2026	Daniel H.	Added problem statement and product name
September 26th, 2026	Arjun	Added custom documents
September 26th, 2026	Noah	Grammar fixes to goals and stretch goals
September 27th, 2026	Peter	Added environment (hardware and software); grammar fixes to inputs, outputs and stakeholders
September 27th, 2026	Team	Call to review documents
September 27th, 2026	Peter	Added reflection section

1 Problem Statement

1.1 Problem

Personal finance and fitness follow the same pattern: the payoff for effort put in today shows up years later. Someone who invests $100 a month from age 25,

assuming a 7% annual return, contributes $48,000 over 40 years and ends up with roughly $262,000. Waiting until 35 to start the same plan leaves them with about $122,000. The same logic applies to exercise. A single walk or workout changes very little, but the habit, kept up for years, is what separates a healthy 40-year-old from an unhealthy one. In both cases the early actions matter most, and in both cases they are the easiest to skip, because the reward is too far away to feel real.

The concepts behind this are not widely understood, and young people understand them the least. A 2012 survey of 6,911 Canadians asked three basic questions covering compound interest, inflation, and risk diversification [1]. Only 42% of respondents answered all three correctly. Among respondents under 35, that number fell to about 30%, the lowest of any age group, and 44% of them answered "don't know" to at least one question. Only about half of respondents under 35 knew that buying a single company's stock is usually riskier than buying a stock mutual fund. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, which suggests that this gap in knowledge carries through to real financial outcomes.

Tools for practising each skill already exist, but separately. Stock market simulators let people trade with fake money, and fitness apps reward steps with points and badges. A market simulator only helps someone who is already motivated to learn about investing, which excludes most of the young adults the survey shows are furthest behind. Fitness rewards, meanwhile, rarely lead to anything beyond the app itself. No product currently connects the two so that a habit a user is already building (or wants to build) becomes their way into learning how money grows.

1.2 Proposed Project

Flexin is a mobile app that turns physical activity into virtual currency, and virtual currency into an investment portfolio. Steps are tracked automatically through the phone's health data rather than entered by hand, and users earn currency by meeting fitness goals such as a daily step count or a completed workout. Users can spend that currency on real financial assets, including stocks, bonds, and ETFs, at a simulated market price. The value of the portfolio rises and falls with the prices of the assets the simulated market holds. Users can view their holdings, their history of purchases and sales, and a graph that separates the currency they contributed from the gains and losses the market produced. Users can also choose to share their fitness and portfolio statistics with the entire community on a leaderboard.

1.3 Inputs and Outputs

1.4 Inputs

1. Taps on the screen: Mobile app users will need to click the screen to complete actions in the application.

2. Activity Information: Connection to a pedometer, heart rate monitor, etc. Many of these technologies can be provided by the device running the application.

3. User Personal information: Information like height, weight, health conditions, current fitness level and current understanding of finance will be entered to enhance the experience.

1.5 Outputs

1. Virtual currency: Users will be able to collect virtual currency by completing fitness tasks and smart investing. Users will then be able to spend the currency as they choose.

2. Real world rewards: virtual currency can be traded for real world incentives (i.e. free trials for paid services).

1.6 Stakeholders

- Users: Young and older adults who would like to learn more about personal fitness and/or financial responsibility.

- Project Supervisor: provided initial project description and will continue to provide insights throughout the project.

- Banks: users who learn more about financials will be more inclined to involve themselves in their personal finances.

- Gyms/Training Facilities: users will be able to learn more about fitness and utilizing gyms/training facilities provides a space for this purpose.

1.7 Environment

Users will be able to use the application from their smart devices. The project will need to be supported by the current hardware and software systems in place that run these smart devices. The application will be developed to be supported by the android operating system and hardware. The distribution of the application will be handled by the respective application stores of the smart devices. Integration with other devices/products that can read fitness data will need to be supported by the smart device. The integration must be supported by an SDK, API, or other package developed by the manufacturer of the device/product that will allow data retrieval.

2 Goals

1. **Educate Users on the Power of Compounding**: Compounding is the process by which financial earnings from investments (dividends and capital gains) are re-invested, accelerating the growth of the investments. It is the most important concept to grasp in order to become financially literate. Fitness is also a compounding habit, where early action can lead to long-term positive outcomes on one's health. The application will educate its users on compounding by rewarding positive fitness habits, as well as healthy financial choices. This can be measured by the number of users who:

   (a) Report a positive learning experience through user surveys

   (b) Show consistent financial growth (dollars gained as a percentage)

   (c) Show consistent workout results through use of app

2. **Integrate with Already Existing Fitness Technologies**: To best track users' fitness activities, the application will integrate with technologies already in use by users. The exact technologies will be determined during the *design thinking* exercise. This can be measured by the number of users who are able to connect at least one of their fitness devices to the application.

3. **Provide a Simple and Delightful User Experience**: The design of the application should be friendly to all users. It should be straightforward, and navigable, with all key components being reachable through minimal clicks. This can be measured through user verification and validation, as well as by following accessibility frameworks.

4. **Drive Long-Term User Engagement**: Compounding only pays off with sustained participation over time. A single workout or investment will not make a difference, but one a week for years will. The application must encourage users to remain active on the platform in the long term, in order to reap the benefits of compounding. This can be measured through the percentage of users who remain active over 30, 60 and 90 days from sign-up.

3 Stretch Goals

1. **Design Unique Interactions for Users with Varying Knowledge Levels**: Create different interactions for users who have more knowledge in finance versus more knowledge in fitness. These unique interactions allow for a more customized learning experience for the users.

2. **Leverage Generative AI in Investments**: Utilize LLMs as a source of learning for the users. The LLMs will be able to provide lessons on both

the fitness side and finance side. Workout and financial advice may also be possible, though regulations will need to be checked.

3. **Incorporate Real Life Incentives**: Partner with companies, brands, and organizations to create in real life (IRL) incentives for the users. These partnerships will drive even more long term engagement with the application as well as further incentivizing proper compounding habits.

The stretch goals listed above are currently out of scope due to time constraints, but may be considered later in the development cycle.

4 Custom Documents (CDs)

We will complete the following two custom documents (one per term):

- **Term 1 (CD-Fall):** User/Stakeholder/Client interview report
- **Term 2 (CD-Winter):** Business Plan

The interview report will contain information that we will use to derive our requirements and goals. The business plan was chosen as we feel this project can potentially become a startup, therefore we want to explore the business side of our project.

References

[1] D. Boisclair, A. Lusardi, and P.-C. Michaud, "Financial literacy and retirement planning in Canada," *Journal of Pension Economics and Finance*, vol. 16, no. 3, pp. 277–296, 2017, doi: 10.1017/S1474747215000311.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   The team worked well to quickly align as we had a similar vision for how we should tackle the problem and create a solution. We also did a good job splitting the work as everyone was eager to participate and motivated to achieve a common goal, to build a product we are proud of.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   The team did at first have some pain points in regards to the direction to take the application. As we are attempting to teach two different principles, personal fitness and personal finance, we needed to ensure that we were focusing on the correct things. We also did have many conversations about the technology stack that we would utilize to complete the application. Based on everyone's own skill sets and previous work/class experience we all had a different idea of the best systems to use. Our group has been organized up to this point and this helped solve these pain points. We have weekly meetings and many lengthy discussions outside of our specific meeting times to ensure we have fully discussed and solved the problems that we have faced prior to continuing.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   For this capstone project we do have a time constraint that we must adhere to. We initially aimed to develop an application that would be able to be used on many operating systems (i.e. IOS, Android, etc.). To ensure that we create the application to the best quality of the team's ability we changed to having it deployed across many platforms to just

focusing on the android development of the application. The deployment to other operating systems has become a stretch goal for the team.